

A comment on “Violence trends in the ancient Middle East between 12,000 and 400 BCE”

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Abstract

In a significant contribution to the quantitative history of the ancient Middle East, Baten et al. (2023) examine long-term trends in interpersonal violence in the ancient Middle East using bioarchaeological evidence from 3,539 skeletons dated from the Mesolithic to the onset of Classical antiquity. Violence is measured as the share of individuals exhibiting cranial trauma or weapon-related injuries, aggregated at the site-period level. Using Tobit regressions with regional fixed effects and contextual controls, the authors report that interpersonal violence peaked during the Chalcolithic period and declined substantially during the Early and especially the Middle Bronze Age. Their specification shows a statistically significant decline in violence during the Middle Bronze Age relative to the Chalcolithic period (log coefficient = -1.815 , $z = -2.55$, $p = 0.011$), which they interpret as an approximate 6.1% reduction in violence prevalence. However, with $b = -1.815$, the usual log interpretation yields $\exp(-1.815) \approx 0.163$, i.e., MBA violence is about 16% of the Chalcolithic level ($\approx 84\%$ lower). The value 6.1 reported in the paper corresponds to $\exp(|b|)$ and is interpretable as a factor ($\approx 6.1\times$ lower), not a 6.1% change. The authors emphasize that these findings are descriptive and robust to controls for urbanization, burial selectivity, climate shocks, population size, and major military innovations.

We successfully computationally reproduce the main results using the authors' analysis data and original Stata code, classifying the study as computationally reproducible from analysis data. However, we document a few inconsistencies between the paper's methodological description and the implemented code, most notably in the treatment of zero violence observations: while the paper describes a $\log(y + 0.05)$ transformation, the estimates are in fact generated using a Tobit specification that treats zero trauma shares as left-censored on the log scale. We further reproduce all main analyses in R and confirm that the key findings hold across software environments. Robustness analyses reveal that the statistical significance and magnitude of some period and contextual coefficients, including the Early Bronze Age effect, are sensitive to the choice of censoring point in the Tobit model, while most remain robust. Finally, we estimate an alternative beta-binomial model that is better aligned with the underlying "count out of n " data-generating process. While this specification preserves the negative association between Middle Bronze Age periods and violence, it yields smaller estimated effects with wider uncertainty and substantially improves predictive calibration, especially for zero-trauma observations. Overall,

while the descriptive decline in violence during the Early-Middle Bronze Age is broadly supported, inference based on the Tobit specification is sensitive to modeling choices that do not fully reflect the bounded, “count out of n” nature of the outcome. Neither the replication nor the sensitivity analyses were pre-registered.

1. Introduction

The data source is a new dataset on cranial trauma and weapon-related wounds from skeletons excavated across seven Middle Eastern countries (Turkey, Iraq, Iran, Syria, Lebanon, Israel and Jordan), spanning the pre-Classical period (around 12,000-400 BCE). The dataset includes more than 3,500 individuals and enables the study of long-term trends in violence in the ancient Middle East.

The study measures interpersonal violence using skeletal evidence, specifically the proportion of individuals showing antemortem or clearly identifiable perimortem cranial trauma or weapon-related injuries. Cranial injuries were limited to areas above the hat-brim line to distinguish violence from accidents, and postmortem damage, stress-related, and accidental fractures were excluded using established osteological criteria. Only individuals aged 15 and older were included. Limited postcranial weapon wounds (for example, arrowheads) were also recorded, though most evidence derives from cranial trauma. Violence intensity was calculated as the share of affected individuals among all sufficiently preserved skeletons at each site or period. The final dataset includes 3,539 individuals, with 323 showing trauma (9.1%). To address potential selection bias, analyses control for urban versus non-urban contexts, special burial practices (such as royal or ritual contexts), and clearly identified battle sites. The authors evaluate whether changes in burial selectivity over time could bias the observed violence trends. Although full representativeness cannot be claimed, the authors argue that deviations are unlikely to fundamentally distort the results. A potential warrior idealization bias (i.e., societies with strong military values might preferentially preserve and excavate warrior burials) is examined in detail. While warrior burials became richer and more common in the Early-Middle Bronze Age, skeletal evidence shows a decline in cranial and weapon-related trauma during the same period, suggesting that such bias does not drive the results and that the actual decline may be even more pronounced; however, it cannot be ruled out that residual selection effects remain. Additional plausibility checks include similar violence trends across different regions, making coordinated

excavation bias less plausible, and consistency with historical evidence linking stronger state institutions, legal systems, and trade expansion to reduced interpersonal violence. Finally, the study formally tests the robustness of these trends using regression analysis with controls for urbanization, climate shocks (droughts), population size, and major military innovations (iron technology and mounted warfare), showing that the decline in violence is not explained by these alternative factors.

The paper's core claims are descriptive with correlational/associational regression analysis. The authors are explicit that they do not claim causality, but they test whether observed temporal changes in violence are robust to a set of theoretically motivated covariates: "We do not make any claims about the direction of causality or the exogenous nature of these variables." (p. 2069). Thus, the correct framing is estimation of interpersonal violence and its association with period and contextual variables, not causal effects. The dependent variable is intensity of interpersonal violence, operationalized as the share of individuals exhibiting cranial trauma or weapon-related wounds, aggregated at the site-period level: "We measure the intensity of interpersonal violence as the share of individuals exhibiting weapon-related wounds or cranial traumas." (p. 2068). The paper tested the association between archaeological period indicators and interpersonal violence for site-period observations in the Ancient Middle East, across regions and time periods, using site-period aggregated bioarchaeological data and multiple Tobit regressions with regional fixed effects and contextual controls. The main results show that violence peaked during the Chalcolithic period and declined substantially during the Early and especially Middle Bronze Age: "Through a standardized methodology for inferring cranial and weapon-related trauma in sufficiently well-preserved skeletons, we find that interpersonal violence peaked during the Chalcolithic period [...] Then violence steadily declined during the EBA-MBA (around 3,300-1,500 bce)." (p. 2068). The primary results demonstrate a statistically significant decline in violence during the Middle Bronze Age relative to the Chalcolithic period, with a log-point estimate of -1.815 ($z = -2.55$, $p = 0.011$): "The coefficient of MBA (relative to Chalcolithic) is $b = -1.815$, $z(70) = -2.55$, $P = 0.011$." (p. 2069). The coefficients are reported in logs, and the authors interpret them as percentage changes: "The coefficients are reported in logs. If we want to express the effect in percentage, we need to take the exponential. For example, the coefficient of MBA ... corresponds to a decline in violence by 6.141%." (p. 2069). So the magnitude E is: $E \approx -6.1\%$ in violence prevalence, relative to the Chalcolithic baseline. However, with $b = -1.815$,

the usual log interpretation yields $\exp(-1.815) \approx 0.163$, i.e., MBA violence is about 16% of the Chalcolithic level ($\approx 84\%$ lower). The value 6.1 reported in the paper corresponds to $\exp(|b|)$ and is interpretable as a factor ($\approx 6.1\times$ lower), not a 6.1% change.

Population: “We study 3,539 well-provenanced skeletons dated to the period from the Mesolithic to the onset of Classical antiquity.” (p. 2070). The population definition for regression is 82 site-period observations across Mesopotamia, Levant, Iran, and Turkey (p. 2069).

Method: Site-period-level aggregation of osteological trauma indicators; multiple Tobit regressions with log-transformed dependent variable; region fixed effects; robust (Huber-White) standard errors clustered at the region-period level: “Seven multiple Tobit regressions show coefficients of various variables potentially related to violence... We use logarithmic transformation; following the standard procedure, we added a small constant (0.05).” (p. 2069).

The following robustness checks are explicitly relevant for re-analysis: Inclusion of contextual controls (Battle sites, royal interments, urban vs rural, climate shocks, iron technology, horse-mounted warfare, population size): “Adding the control variables one by one, we observe that the coefficients for EBA-MBA are almost unchanged.” (p. 2069) The authors name alternative model specifications: “We tested a model without controls and then added the controls one by one.” (p. 2069). Regarding sensitivity analyses, the authors write “We perform a series of tests in Supplementary robustness and sensitivity tests [...] including interaction terms, alternative modeling strategies, and anthropological sensitivity analyses.” (p. 2069). Regarding small-N exclusion checks, the authors explain “We also performed a robustness test [...] in which we include or exclude cases with low numbers of observations.” (p. 2067).

The authors frame the decline of interpersonal violence during the Early-Middle Bronze Age (EBA-MBA) as the central finding of the study: “Through a standardized methodology for inferring cranial and weapon-related trauma in sufficiently well-preserved skeletons, we find that interpersonal violence peaked during the Chalcolithic period [...]. Then violence steadily declined during the EBA-MBA (around 3,300-1,500 bce).” (p. 2068). Our replication target and the main claim is therefore: Interpersonal violence in the Ancient Middle East declined substantially during the Early and especially Middle Bronze Age relative to earlier periods, and

this decline is robust to controls for region, settlement type, climate shocks, military technologies, and population size.

2. Computational Reproducibility

We obtained the data via links in the online version of the original article (<https://zenodo.org/records/8010025>). We did not contact the authors with any requests or questions. No raw data and no cleaning code are provided by the authors. The analysis data is fully provided, but the code is only partially provided. Since the authors do not provide raw data, the results are not reproducible from raw data but are fully reproducible from the analysis data (see Table 1).

We first ran the original script on the original study data in Stata, the coding environment that the authors have used. We have found some minor reporting issues:

- The code for Figure 4 in the paper is not provided
- The authors report z-statistics with degrees of freedom notation (e.g., " $z(70) = -4.33$ "), but z-statistics do not have degrees of freedom as they follow the standard normal distribution. The Stata output does not show any degrees of freedom or corresponding numbers, suggesting that this is a reporting error where the authors manually calculated and added degrees of freedom notation to z-statistics.

We then proceeded to check the robustness of the analysis. We used the original data and translated the original Stata script into R. We did not use any new data for the replication.

2.1 Reproducibility of the main findings

We successfully computationally reproduced the main results from the analysis data. This classifies the paper as “Computationally Reproducible from Analysis Data: analysis code with analysis data produces the same output as presented in the paper”. The replication code is provided on GitHub (<https://github.com/Pascal-Kueng/Replicating-Violence>).

2.2 Discrepancies between the reported and implemented analysis

The paper states that the dependent variable is log-transformed after adding a small constant (0.05) to accommodate zero trauma shares. In the original estimation code we examined, we do not find a '+0.05' transformation step. Accordingly, reproducing the published results does not require (and is not consistent with) a $\log(y + 0.05)$ specification.

Instead, the published estimates are exactly reproduced when the outcome is constructed as follows: the log is taken only for strictly positive trauma shares, and observations with a trauma share of zero are not transformed into a finite log value but are treated as left-censored at a fixed censoring point on the log scale (approximately -3). This approach differs from adding a constant before logging. Under the implemented approach, zeros contribute to the likelihood through a censoring assumption ("latent below the threshold") rather than through an observed transformed value. Because this censoring step is central to the model's interpretation and to how zeros are handled, the mismatch between the paper's description and the implemented code is not trivial.

3. Robustness Reproduction

The authors investigated the robustness of their findings through various sensitivity analyses. They included and excluded time-region units with small numbers of observations, included the right-hand-side variables separately in the model rather than cumulating them, included an interaction term for urban place and period, and included or excluded the parry fractures.

We now further stressed the robustness of the finding by running additional robustness checks. First, we reproduced the main findings using R rather than Stata. Second, we varied the censoring points in the Tobit regression. Third, we run an alternative regression using a beta-binomial specification. In our robustness analyses, we focused on Model 3 because it is the main model discussed in the article.

3.1. Replication in R

We were able to reproduce the original findings computed in Stata using R. We replicated the main analyses in R (v.4.5.2) using base R and a custom likelihood function for the Tobit models, and the package glmmTMB (v.1.1.14) for the beta-binomial model. DHARMA (v.0.4.7) was used for residual analysis. The findings are provided in Table 2.

3.2. Sensitivity of the Tobit results to the censoring point

The Tobit specification appears to be a pragmatic strategy to address the “log of zero” problem inherent in sparse data. However, because the Tobit model assumes an unbounded underlying distribution, it does not naturally constrain predictions to the $[0, 1]$ interval required for proportion data. The trauma measure is a proportion derived from the number of trauma cases observed out of the number of examined skeletons in a site-period. In this setting, a zero share typically means “0 cases out of n examined,” not that an underlying continuous latent variable has been censored by a detection limit. Tobit’s key assumption (i.e., that there exists an unbounded latent normal outcome that can fall below a threshold and then becomes censored) does not naturally correspond to a “count out of n ” data-generating process.

A related concern is that the information content of a zero depends heavily on the denominator. Observing 0 cases out of a large sample provides much stronger evidence that the underlying rate is near zero than observing 0 cases out of a small sample. In the authors’ Tobit setup, zeros are handled via a fixed censoring point (roughly equivalent to a 5% violence rate). In this model, both 0/5 and 0/500 enter as “left-censored below the same threshold”. Weights can change influence, but the likelihood still does not reflect the binomial sampling evidence that makes 0/500 far more informative than 0/5.

The Tobit specification requires choosing a censoring point on the log scale (the paper’s implementation corresponds to approximately $c \approx \log(0.05) \approx -3.0$). Because this threshold is not implied by the underlying “count out of n ” data-generating process, we re-estimated the full Tobit model across a grid of plausible censoring values $c \in \{-5.0, -4.5, -4.0, -3.5, -3.0, \text{ and } -2.5\}$, holding the covariates, analytic weights, clustering, and estimation procedure fixed.

Table 2 shows that some effects shift as c changes. For example, the estimate for the Early Bronze Age is not statistically significant at very low censor points ($c = -5.0$ and -4.5) but becomes significant for higher thresholds (from $c = -4.0$ onward). There are some changes in the covariates. For example, the coefficient for region is statistically supported at lower censor points but loses support as the threshold rises (from roughly $c = -3.5$ onward). Substantively, key context coefficients such as Battle Sites and Royal Interments also shrink as c increases, indicating that effect magnitudes are not completely stable to this modeling choice.

Within the Tobit likelihood, model fit improves mechanically as it increases (logLik rises and AIC/BIC fall; see Table 3). However, this improvement coincides with sharply worsening calibration of zeros: while the observed weighted censoring rate is 0.125, the model's predicted weighted censoring probability rises from 0.126 at $c = -5.0$ to 0.600 at $c = -3.0$ and 0.711 at $c = -2.5$. Thus, the censor points that appear “best” by AIC/BIC correspond to the most inconsistent with the observed frequency of zeros and exacerbate the model's misalignment with the bounded outcome.

The Tobit results, and in particular which period and proxy effects appear statistically supported, are sensitive to an analyst's chosen censoring threshold. This sensitivity reinforces that the Tobit/censoring formulation can be a fragile basis for inference in a “count-out-of- n ” setting.

3.3. Comparing Model Fit and Predictive Accuracy to a Beta Binomial Specification

Another concern was that the proportional/count outcomes have a mean-variance relationship dictated by binomial sampling (and often additional overdispersion). The Tobit framework assumes (on the latent scale) a constant-variance normal disturbance; analytic weights can rescale contributions to the likelihood, but they do not make the Tobit likelihood equivalent to the count-based sampling process that generated the data, nor do they automatically recover the correct dependence of variance on both n and the mean.

A beta-binomial model matches the structure of the data directly. It models the observed number of trauma cases k out of n examined individuals, with the probability of trauma π linked to covariates (e.g., via a logit link). The binomial component handles the fact that the data are discrete counts bounded by n ; the beta component allows π to vary across site-periods beyond

what a strict binomial would permit, capturing overdispersion that is plausible in aggregated bioarchaeological data.

This approach addresses the key conceptual issues above:

- Zeros are treated as ordinary outcomes with an exact probability $P(k = 0)$, rather than as censored latent values requiring an arbitrary censor threshold.
- Differences in denominators are handled natively through the likelihood: 0/500 and 0/10 are not treated as equivalent signals.
- Predicted means remain within $[0, 1]$ by construction.
- Overdispersion can be modeled explicitly, which generally yields more credible uncertainty estimates than relying on a homoskedastic latent-normal assumption.

The analysis sample consists of $N = 84$ site-period observations (34 zero-trauma cases). To implement the beta-binomial model, continuous proportions were reconstructed into integer counts, ensuring the approximation is small (mean absolute discrepancy 0.021 individuals), so the reconstructed counts closely track the original shares.

While the Tobit model technically reports a lower AIC (234.7 vs 329.5), these metrics are not comparable because the models operate on different outcome scales (continuous log-density vs. discrete counts). Instead, we rely on “common-scale” predictive checks, which reveal substantial miscalibration in the Tobit specification:

- Predictive checks (back-transformed means and zero predictions): The Tobit model generates predictions outside the possible range $[0, 1]$ when projected back to the original scale. Because the latent normal is unbounded, $\exp(y^*)$ can be arbitrarily large; with heteroskedasticity from weights, this yields extreme means. This is a property of exponentiating an unbounded normal index (lognormal moments), not a feature of the bounded data. The weighted predicted mean trauma share for the Tobit model was 2,330.16, compared to an observed weighted mean of 0.091. This issue is systematic across periods: for example, the Tobit model predicts weighted means of 5,677 for the period Early Bronze Age and 7,896 for Middle Bronze Age, against observed means of < 0.100 . This occurs because the Tobit model applies analytic weights to the log-scale

variance; when back-transformed to the natural scale, this inflated variance creates a heavy right tail that does not exist in the bounded $[0, 1]$ data.

- Zero predictions: The Tobit model substantially overpredicts the frequency of zeros. It predicts a weighted zero rate of 59.9%, whereas the observed weighted zero rate is only 12.6%. The beta-binomial model closely matches the data structure, with a predicted zero rate of 16.4%.
- Residual Diagnostics: DHARMA simulation-based diagnostics did not indicate major departures from expected residual behavior (KS test $p = 0.59$) and did not suggest excess zeros beyond the fitted model ($p = 0.58$).

4. Conclusion

The most important result of our work is that the central descriptive finding of the original study, a substantial decline in interpersonal violence during the Early and especially Middle Bronze Age relative to earlier periods, can be computationally reproduced using the authors' analysis data and code, and is qualitatively robust across software environments. All our robustness checks agree that Battle Sites and Royal Interments contexts are associated with significantly higher trauma prevalence, and all specifications consistently indicate that the Middle Bronze Age period has significantly lower trauma prevalence than the baseline period, although the effect is smaller and accompanied by wider uncertainty in the beta-binomial model. Overall, trauma prevalence is higher in Battle Sites and Royal Interments contexts and lower in the Middle Bronze Age. At the same time, our analysis reveals that key aspects of the empirical strategy are more sensitive than the original paper may imply. Most importantly, we identify a discrepancy between the paper's methodological description and the implemented analysis regarding the treatment of zero trauma observations. While the paper describes a log transformation with an added constant, the estimates are in fact generated by a Tobit model that treats zero shares as left-censored outcomes on the log scale. This modeling choice is not a trivial detail, as varying the censoring point materially affects the magnitude and statistical significance of several coefficients, including those for the Early Bronze Age and some contextual covariates. These sensitivities indicate that inference within the Tobit framework depends heavily on analyst's

chosen assumptions that are not directly implied by the underlying bioarchaeological data-generating process.

In the original Tobit specification, three additional periods, the Neolithic, Early Bronze Age, and Iron Age, are identified as having significantly lower trauma rates relative to the Chalcolithic peak. The beta binomial model finds no evidence for these effects, suggesting that they may be partly driven by the Tobit censoring specification, which can treat small-sample zeros as high-certainty data points. These period differences were not robust to a count model with overdispersion and appear sensitive to the Tobit censoring specification, particularly because we did not reproduce direct comparisons between non-baseline periods.

We further see that modeling interpersonal violence as a bounded count process rather than as a censored latent continuous variable substantially improves empirical coherence and predictive performance. A beta binomial specification, which directly models the number of trauma cases out of the number of examined individuals, naturally accommodates zeros, differences in denominators, and overdispersion. While this alternative model continues to support a negative association between the Middle Bronze Age period and violence, the estimated effects are smaller and less precisely measured, and the resulting predictions are well calibrated and remain within plausible bounds. This suggests that the headline pattern in the data is supported, but that the strength and statistical certainty of the decline depend on the chosen modeling framework.

Overall, this replication highlights the importance of transparency in data construction and outcome transformation, especially when dealing with proportional and zero-heavy data. Small differences between stated and implemented methods can have first-order consequences for interpretation, even when headline results appear stable. More generally, our results point to the benefits of count-based models for aggregated trauma data. We emphasize that the paper's main contributions, including the assembly and analysis of a large, well-documented trauma dataset over very long time horizons, remains substantial.

References

Baten, J., Benati, G., & Sołtysiak, A. (2023). Violence trends in the ancient Middle East between 12,000 and 400 BCE. *Nature human behaviour*, 7(12), 2064-2073.

Appendix

Table 1: Provision of data and analysis code of the authors, and reproducibility.

	Fully	Partial	No
Raw data provided			x
Cleaning code provided			x
Analysis data provided	x		
Analysis code provided		x	
Reproducible from raw data			x
Reproducible from analysis data	x		

Table 2: Comparison of the results of the main study and the robustness checks.

Variables	Original (0)	Replication in R (1)	Different censoring points (2)						Beta binomial model (3)
			-5.0	-4.5	-4.0	-3.5	-3.0 (original)	-2.5	
	Estimate (SE)	Estimate (SE)	Estimate (SE)	Estimate (SE)	Estimate (SE)	Estimate (SE)	Estimate (SE)	Estimate (SE)	Log-odds (SE)
Neolithic	-1.332* (0.308)	-1.332* (0.308)	-1.085* (0.308)	-1.150* (0.308)	-1.213* (0.308)	-1.274* (0.308)	-1.332* (0.308)	-1.386* (0.308)	-0.303 (0.473)
Early Bronze Age	-0.800* (0.249)	-0.800* (0.249)	-0.438 (0.249)	-0.534 (0.249)	-0.628* (0.249)	-0.717* (0.249)	-0.800* (0.249)	-0.877* (0.249)	-0.077 (0.416)
Middle Bronze Age	-1.815* (0.712)	-1.815* (0.712)	-2.572* (0.712)	-2.366* (0.712)	-2.170* (0.712)	-1.986* (0.712)	-1.815* (0.712)	-1.661* (0.712)	-1.278* (0.624)
Late Bronze Age	0.715 (0.810)	0.715 (0.810)	1.886 (0.810)	1.567 (0.810)	1.263 (0.810)	0.978 (0.810)	0.715 (0.810)	0.479 (0.810)	0.645 (0.435)
Iron Age	-1.114* (0.254)	-1.114* (0.254)	-1.316* (0.254)	-1.263* (0.254)	-1.211* (0.254)	-1.161* (0.254)	-1.114* (0.254)	-1.068* (0.254)	-0.242 (0.555)
Battle sites	1.157* (0.446)	1.157* (0.446)	2.071* (0.446)	1.824* (0.446)	1.588* (0.446)	1.365* (0.446)	1.157* (0.446)	0.968* (0.446)	1.414* (0.462)
Royal interments	1.861* (0.469)	1.861* (0.469)	2.632* (0.469)	2.421* (0.469)	2.220* (0.469)	2.033* (0.469)	1.861* (0.469)	1.708* (0.469)	1.308* (0.257)
Urban	0.551	0.551	0.533	0.537	0.542	0.546	0.551	0.556	0.096

	(0.429)	(0.429)	(0.429)	(0.429)	(0.429)	(0.429)	(0.429)	(0.429)	(0.390)
Constant	-1.939* (0.451)	-1.939* (0.451)	-2.031* (0.451)	-2.004* (0.451)	-1.980* (0.451)	-1.958* (0.451)	-1.939* (0.451)	-1.925* (0.451)	-1.544* (0.730)

Note: Table comparing the original estimates from the main model (Model 3) in the analyses with the robustness checks (differences in bold).

Table 3: Comparison of the fit for different censoring points.

cpoint	logLik	AIC	BIC	obs_cens_w	pred_cens_w
-5.0	-124.957	275.914	307.201	0.125	0.126
-4.5	-119.917	265.834	297.121	0.125	0.159
-4.0	-114.720	255.439	286.727	0.125	0.258
-3.5	-109.478	244.957	276.244	0.125	0.441
-3.0	-104.356	234.713	266.000	0.125	0.600
-2.5	-99.551	225.101	256.389	0.125	0.711

Note: Each row reports the Tobit/intreg-style model re-estimated with an alternative censoring point (c) (log scale), holding covariates, analytic weights, clustering, and estimation identical. (obs_cens_w) is the observed weighted share of left-censored observations (zeros), and (pred_cens_w) is the model-implied weighted probability of censoring averaged over observations.